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Algorithmen und Wahrscheinlichkeit
Exercise Session 6

Exercise 1 – *Probabilistic method*

This is one of the most beautiful applications of so called "Probabilistic Method", due to Paul Erdős — a method of using probability theory to show existence of specific combinatorial objects, that on its face have nothing to do with probability.

Specifically, as mentioned in Lecture 7, we will show that every k, there is a graph G that has no triangle (so the largest complete subgraph is just K_2), but has chromatic number at least k.

Let us consider a random graph on n vertices, such that every edge is included independently with probability $p = \alpha/n$, where $\alpha = \log n$, and let $\beta = m/n$ for some $m < n$ be a small constant (it will be chosen s.t. $\beta \approx 1/2k$).

(a) Show that if T is a random variable denoting the number of triangles in G, then $\mathbb{E}[T] \leq \alpha^3$.

(b) Let M be a random variable denoting the number of independent sets of size $n\beta$ in G. Show that

$$\mathbb{E}[M] \leq 2^n \exp(-\alpha\beta^2 n/3).$$

Conclude that when n is large enough, probability that G has an independent set of size $n\beta$ is at most $\exp(-100)$.

(c) Show that this implies that for every n larger than some n_0, there is a graph G with at most $10\alpha^3 = 10\log^3 n$ triangles, and no independent set of size $n\beta$.

(d) Show if n is large enough, this implies that there is a graph G' with no triangles, and chromatic number at least $2/\beta$.

Solution 1

(a) For any three distinct vertices $x, y, z \in [n]$, Let $T_{x,y,z}$ be an indicator random variable denoting whether triangle $\{x, y, z\}$ is present in G (i.e. all three edges $\{x, y\}, \{y, z\}$ and $\{z, x\}$ are present in G). On one hand, since the edges are independent, we have $\mathcal{E}T_{x,y,z} = p^3$, and therefore

$$\mathcal{E}[T] = \sum_{(x,y,z)\in\binom{[n]}{3}} \mathcal{E}[T_{x,y,z}] = \binom{n}{3}p^3 \leq n^3p^3 \leq \alpha^3.$$

(b) For a given subset $U \subset [n]$ of size βn, let M_U be an indicator random variable denoting that U is independent set in G. For n large enough we have $\binom{\beta n}{2} = \frac{\beta n(\beta n-1)}{2} \leq \beta^2 n^3/3$. Therefore

$$\mathcal{E}[M_U] = (1-p)^{\binom{\beta n}{2}} \leq (1-p)^{\beta^2 n^2/3} \leq \exp(-pn^2\beta^2/3) \leq \exp(-n\alpha\beta^2/3).$$

Since $M = \sum_{U\in\binom{n}{\beta n}} M_U$ and $\binom{n}{\beta n} \leq 2^n$ (the number of subsets of size βn is at most the number of all subsets, we have

$$\mathcal{E}M \leq 2^n \exp(-n\alpha\beta^2/3) = \exp(-n\alpha\beta^2/3 + n\ln 2).$$

If $\alpha = \log n$ and β is a small constant, then for n large enough, we have $-n\log n\beta^2/3 + n\ln(2) < -100$, and hence $\mathcal{E}[M] < \exp(-100)$. By Markov inequality,since $M \geq 0$, we have $\Pr[M > 1] \leq \mathcal{E}[M] \leq \exp(-100)$.

(c) Let A be an event that G has more than $3\alpha^3$ triangles, and B be an event that G has independent set of size n. By union bound, we have $\Pr[A \cup B] \leq \Pr[A] + \Pr[B]$, and by Markov inequality $\Pr[A] \leq 1/3$. By previous exercise, $\Pr[B] \leq \exp(-100)$. Hence $\Pr[A \cup B] \leq 1/3 + \exp(-100) \leq 1/2$. So there exist a graph for which neither A nor B holds.

(d) Consider a graph G' where we take graph G as in the previous sub-exercise and remove all vertices in all triangles in G. We remove at most $3\alpha^3 = 3\log^3 n$ vertices, so if n is large enough, $3\log^3 n < n/2$, and the graph G' has $n' > n/2$ vertices. Clearly, by construction G' is triangle free, and moreover has no independent set of size βn. As such, the chromatic number of G' is at least $n'/(\beta n) \geq n/(2\beta n) \approx (1/2\beta)$.

Exercise 2 – *Extra exercise*

Adjust those ideas to show that for every t and k there is a graph with no cycles shorter than t, and chromatic number at least k.